A Hilbert–Schmidt Scaling Limit for Finite-Interval Wilson–Dirac Covariances

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Abstract. For the restrictions of two infinite-chain Wilson–Dirac occupied-band projections to an interval of L cells, we establish a Hilbert–Schmidt scaling limit when the massive parameter is $m = \mu/L$ with fixed $\mu > 0$. The limiting integral operator on $L^2(0,1;\mathbb{C}^2)$ has an explicitly computable modified-Bessel kernel. A dimension-free lower bound for the finite quasifree relative entropy then gives a strictly positive limiting lower benchmark, without a spectral-mode localization hypothesis. The benchmark is an increasing, strictly convex function of μ; its small-mass expansion and large-mass slope are explicit. The arguments use only the occupied-band symbol, Fourier–Plancherel theory, and finite-matrix convexity.

Keywords: Wilson–Dirac chain; quasifree fermions; Hilbert–Schmidt convergence; quantum relative entropy; Bessel functions; Toeplitz compression.

1. Model and result

Write σ_x, σ_z for the Pauli matrices and fix $\mu > 0$. For $m \geq 0$ and $k \in [-\pi, \pi]$, put

$$h_m(k) = \sin k\, \sigma_x + (1 + m - \cos k)\, \sigma_z, \qquad E_m(k) = \sqrt{m^2 + 4(1+m)\sin^2(k/2)}.$$

The infinite-chain occupied-band multiplier is $P_m(k) = \frac{1}{2}(I - h_m(k)/E_m(k))$. Its value at $k = m = 0$ may be specified arbitrarily, since a single Fourier point is immaterial. Let $C_{L,m}$ be the block Toeplitz compression to L consecutive two-orbital cells,

$$(C_{L,m})_{jk} = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(j-k)\theta} P_m(\theta)\, d\theta, \qquad 0 \leq j, k < L.$$

We compare the massive and massless compressions at $m = \mu/L$:

$$A_L = C_{L,\mu/L}, \qquad B_L = C_{L,0}, \qquad \Delta_L = A_L - B_L.$$

The finite compressed covariances satisfy $0 < C_{L,m} < I$ for every $m \geq 0$. To see this, suppose that a nonzero finitely supported vector has zero quadratic form under $C_{L,m}$ or $I - C_{L,m}$. Its nonzero vector-valued Laurent polynomial $v(z)$, with $z = e^{ik}$, would then satisfy $h_m(k)v(z) = \pm E_m(k)v(z)$ almost everywhere. A nonzero component forces $E_m(k)$ to agree there with a rational function of z. But

$$E_m(k)^2 = m^2 + 2(1+m) - (1+m)(z + z^{-1})$$

has a pole of odd order at $z = 0$, whereas the square of a rational function has only poles of even order. This contradiction proves both strict inequalities.

The trace convention $\|M\|_2^2 = \mathrm{Tr}(M^*M)$ is **un-normalized**, and the logarithms in the relative entropy are natural. Define

$$\mathcal{D}_L(\mu) = \mathrm{Tr}\{A_L(\log A_L - \log B_L) + (I - A_L)[\log(I - A_L) - \log(I - B_L)]\}.$$

This is the relative entropy of the corresponding finite gauge-invariant quasifree states.

Theorem 1 (Covariance scaling and relative-entropy bound). For fixed $\mu > 0$, the matrices Δ_L have a Hilbert–Schmidt continuum limit after the canonical piecewise-constant embedding of the L cells into $(0,1)$. Its

kernel for $s = x - y \neq 0$ is

$$\mathcal{K}_\mu(s) = \frac{i}{2\pi}\left[\frac{1}{s} - \mu\,\mathrm{sgn}(s)K_1(\mu|s|)\right]\sigma_x - \frac{\mu}{2\pi}K_0(\mu|s|)\sigma_z. \tag{1}$$

In particular,

$$\boxed{\lim_{L\to\infty}\|\Delta_L\|_2^2 = H(\mu) = \frac{1}{\pi^2}\int_0^1 (1-s)\left\{\mu^2 K_0(\mu s)^2 + \left[\frac{1}{s} - \mu K_1(\mu s)\right]^2\right\} ds.} \tag{2}$$

The associated relative entropy obeys

$$\boxed{\liminf_{L\to\infty}\mathcal{D}_L(\mu) \geq 2H(\mu) > 0.} \tag{3}$$

The additional estimates in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* establish the relative entropy's leading logarithmic coefficient with bounded remainder. Equation (3) is a separate all-mass, fixed-μ lower benchmark that can be evaluated without using that theorem.

2. Proof of the Hilbert–Schmidt limit

Lemma 2 (Rescaled Fourier multipliers). Set

$$F_L(t) = \mathbf{1}_{[-\pi L,\pi L]}(t)[P_{\mu/L}(t/L) - P_0(t/L)].$$

As $L\to\infty$, $F_L \to F_\mu$ in $L^2(\mathbb{R}; M_2(\mathbb{C}))$, where, for $t \neq 0$,

$$F_\mu(t) = -\frac{1}{2}\left\{\left[\frac{t}{\sqrt{t^2+\mu^2}} - \mathrm{sgn}\,t\right]\sigma_x + \frac{\mu}{\sqrt{t^2+\mu^2}}\sigma_z\right\}. \tag{4}$$

Moreover $\|tF_L\|^2_{L^2(\mathbb{R};\mathrm{F})} = O_\mu(L)$.

Proof. For fixed $t \neq 0$, use $L\sin(t/L) \to t$ and $L(1-\cos(t/L)) \to 0$ in the normalized Bloch vector. The massless σ_x component tends to $\mathrm{sgn}\,t$, while its σ_z component tends to zero, giving (4).

For $k \neq 0$ in the Brillouin zone, the vector $h_m(k)$ has length $E_m(k) \geq E_0(k) = 2|\sin(k/2)| \geq 2|k|/\pi$. Differentiating its normalized direction with respect to m gives a Euclidean norm at most $1/E_m(k)$. The mean-value bound for the difference of the two rank-one spectral projections, supplemented by the elementary uniform bound for differences of projections, therefore gives

$$\|F_L(t)\|_\mathrm{F} \leq C\min\left\{1, \frac{\mu}{|t|}\right\} \quad (t \neq 0), \tag{5}$$

with C independent of L and μ. The right side is square-integrable in t for fixed $\mu > 0$ and also bounds F_μ. Dominated convergence proves the L^2 assertion. Finally $|t|\|F_L(t)\|_\mathrm{F} \leq C\mu$ on its support, an interval of length $2\pi L$, so the stated weighted L^2 bound follows. □

Let

$$K_L(s) = \frac{1}{2\pi}\int_\mathbb{R} e^{its}F_L(t)\,dt, \qquad K_\mu = \mathcal{F}^{-1}F_\mu$$

using the same Fourier normalization. Plancherel and Lemma 2 imply

$$\|K_L - K_\mu\|_{L^2(\mathbb{R};\mathrm{F})} \longrightarrow 0, \qquad \|K_L'\|^2_{L^2(\mathbb{R};\mathrm{F})} = O_\mu(L). \tag{6}$$

In particular $K_L \in H^1(\mathbb{R}; M_2)$ has a continuous representative, so its lattice samples are well-defined.

Lemma 3 (Sampling does not change the limit). Map the standard basis of $\mathbb{C}^L$ into $L^2(0,1)$ by $e_j \mapsto \sqrt{L}\,\mathbf{1}_{[j/L,(j+1)/L)}$. Under its tensor product with $I_{\mathbb{C}^2}$, the finite covariance difference becomes an integral operator with kernel

$$M_L(x,y) = K_L\left(\frac{\lfloor Lx\rfloor - \lfloor Ly\rfloor}{L}\right).$$

Then $\|M_L - K_\mu(x-y)\|_{L^2((0,1)^2;\mathrm{F})} \to 0$.

Proof. The exact block Fourier formula, after $t = L\theta$, gives $(\Delta_L)_{jk} = L^{-1}K_L((j-k)/L)$, which establishes the displayed kernel. If x and y belong to the jth and kth cells, let $s_0 = (j-k)/L$ and $s = x-y$. Then $|s-s_0| \le 1/L$, and the fundamental theorem of calculus and Cauchy–Schwarz give

$$\|K_L(s) - K_L(s_0)\|_{\mathrm{F}}^2 \le \frac{1}{L}\int_{s_0-1/L}^{s_0+1/L} \|K_L'(u)\|_{\mathrm{F}}^2 du.$$

Each pair of cells has area L^{-2}; for each difference $d = j-k$, there are $L-|d|$ pairs, and the intervals $[(d-1)/L, (d+1)/L]$ overlap at most threefold. Consequently,

$$\|M_L - K_L(x-y)\|_{L^2((0,1)^2;\mathrm{F})}^2 \le \frac{3}{L^2}\|K_L'\|_{L^2(\mathbb{R};\mathrm{F})}^2 = O_\mu(L^{-1}).$$

Also

$$\int_0^1\int_0^1 \|K_L(x-y) - K_\mu(x-y)\|_{\mathrm{F}}^2 dx\,dy = \int_{-1}^1 (1-|s|)\|K_L(s) - K_\mu(s)\|_{\mathrm{F}}^2 ds \le \|K_L - K_\mu\|_{L^2(\mathbb{R};\mathrm{F})}^2.$$

Equation (6) completes the proof. □

Lemma 4 (Modified-Bessel kernel). For $s \neq 0$, $K_\mu(s)$ is given by (1).

Proof. The cosine transform

$$\int_0^\infty \frac{\cos(ts)}{\sqrt{t^2+\mu^2}} dt = K_0(\mu|s|)$$

and $K_0'(z) = -K_1(z)$ give the Fourier transform of $t/\sqrt{t^2+\mu^2}$ by differentiation. The Fourier transform of $\operatorname{sgn} t$ is $i/(\pi s)$ in the tempered-distribution sense with the present sign convention. Their difference lies in L^2; the singular terms cancel when combined. The σ_z component is the ordinary even cosine transform. Substitution into (4) yields (1). □

Proof of the limit (2). The $K_0(\mu|s|)$ component has a logarithmic singularity at $s = 0$; the coefficient of σ_x vanishes there at rate $O_\mu(s|\log|s||)$. Thus the kernel is locally square-integrable. Orthogonality of the Pauli matrices under the Frobenius inner product yields

$$\|K_\mu(s)\|_{\mathrm{F}}^2 = \frac{1}{2\pi^2}\left\{\left[\frac{1}{s} - \mu\operatorname{sgn}(s)K_1(\mu|s|)\right]^2 + \mu^2 K_0(\mu|s|)^2\right\}.$$

Lemma 3 gives convergence of the embedded operators in Hilbert–Schmidt norm. Integrating over the unit square, using $\int_0^1\int_0^1 f(x-y)dxdy = \int_{-1}^1 (1-|s|)f(s)ds$, proves (2). □

3. Finite-dimensional entropy inequality

Proposition 5 (Sharp covariance Hilbert–Schmidt bound). If $0 < A, B < I$ are finite Hermitian matrices of the same size, then

$$\mathcal{D}(A\|B) \ge 2\operatorname{Tr}(A-B)^2. \tag{7}$$

The coefficient 2 cannot be increased uniformly over all such matrices.

Proof. Let $f(u) = u\log u + (1-u)\log(1-u)$ on $(0,1)$, and put $V = A - B$, $B_t = B + tV$, and $g(t) = \operatorname{Tr} f(B_t)$. Direct rearrangement of the trace entropy gives $\mathcal{D}(A\|B) = g(1) - g(0) - g'(0)$. Diagonalize B_t with eigenvalues $\lambda_i(t)$. Differentiation of a trace spectral function gives

$$g''(t) = \sum_{i,j} |V_{ij}^{(t)}|^2 \begin{cases} \dfrac{f'(\lambda_i) - f'(\lambda_j)}{\lambda_i - \lambda_j}, & \lambda_i \neq \lambda_j, \\ f''(\lambda_i), & \lambda_i = \lambda_j. \end{cases}$$

This divided-difference identity is valid also at degeneracies by continuous extension. Since $f''(u) = 1/[u(1-u)] \geq 4$, every coefficient in the sum is at least 4 by the scalar mean-value theorem. Therefore $g''(t) \geq 4\|V\|_2^2$. Taylor's integral identity gives

$$\mathcal{D}(A\|B) = \int_0^1 (1-t)g''(t)dt \geq 2\|V\|_2^2.$$

For one-dimensional $B = 1/2$ and $A = 1/2 + \epsilon$, the ratio $\mathcal{D}(A\|B)/\epsilon^2 \to 2$ as $\epsilon \to 0$, proving sharpness. □

Proof of (3). Apply (7) to A_L and B_L, and take the lower limit using (2). The strict positivity follows from $K_0(t) > 0$ for $t > 0$, since its squared contribution has positive integral. □

4. Dependence on the mass parameter

Put

$$G(t) = K_0(t)^2 + \left(K_1(t) - \frac{1}{t}\right)^2, \qquad t > 0.$$

The change of variable $t = \mu s$ gives a useful second representation,

$$\boxed{H(\mu) = \frac{1}{\pi^2}\int_0^\mu (\mu - t)G(t)\,dt.} \tag{8}$$

The behavior of K_0 and K_1 at zero shows $G(t) = O(1 + |\log t|^2)$, while $G(t) = t^{-2} + O(e^{-t}t^{-1/2})$ at infinity.

Corollary 6 (Monotonicity and convexity). The function H is twice continuously differentiable on $(0, \infty)$ and

$$H'(\mu) = \frac{1}{\pi^2}\int_0^\mu G(t)dt > 0, \qquad H''(\mu) = \frac{G(\mu)}{\pi^2} > 0. \tag{9}$$

Proposition 7 (Exact integral and large-mass slope). One has

$$\int_0^\infty \left[K_0(t)^2 + \left(K_1(t) - \frac{1}{t}\right)^2\right] dt = \pi, \qquad \boxed{\lim_{\mu\to\infty} \frac{H(\mu)}{\mu} = \frac{1}{\pi}.} \tag{10}$$

Proof. Formula (4), with $\mu = 1$, gives the exact Frobenius square

$$\|F_1(t)\|_{\mathrm{F}}^2 = 1 - \frac{|t|}{\sqrt{t^2+1}}.$$

Its integral over $\mathbb{R}$ is 2, by the elementary antiderivative $t - \sqrt{1+t^2}$ on the positive axis. Fourier–Plancherel thus gives $\int_{\mathbb{R}} \|K_1(s)\|_{\mathrm{F}}^2 ds = 1/\pi$. Inserting the Bessel kernel (1) and using its even Frobenius norm gives the first identity in (10). The expression $H(\mu)/\mu = \pi^{-2}\int_0^\mu (1 - t/\mu)G(t)dt$ converges to $\pi^{-2}\int_0^\infty G(t)dt$ by dominated convergence. □

Proposition 8 (Small-mass expansion). Let $a_\mu = \log(2/\mu) - \gamma$, where γ is Euler's constant. As $\mu \downarrow 0$,

$$\boxed{H(\mu) = \frac{\mu^2}{\pi^2}\left(\frac{a_\mu^2}{2} + \frac{3a_\mu}{2} + \frac{7}{4}\right) + O(\mu^4[1 + |\log\mu|]^2)\,.} \tag{11}$$

Proof. The standard modified-Bessel expansions, uniformly for $0 < s \leq 1$ and small μ, give

$$K_0(\mu s) = a_\mu - \log s + O(\mu^2 s^2[1 + |\log(\mu s)|])\,, \quad \frac{1}{s} - \mu K_1(\mu s) = O(\mu^2 s[1 + |\log(\mu s)|])\,.$$

Use these in (2). The integrable error terms have the stated order. The elementary integrals $\int_0^1 (1-s)ds = 1/2$, $\int_0^1 (1-s)(-\log s)ds = 3/4$, and $\int_0^1 (1-s)(\log s)^2 ds = 7/4$ yield (11). □

The limit $L \to \infty$ throughout this manuscript is taken with μ fixed. Equations (10)–(11) describe the **subsequent** behavior of the limiting function $H(\mu)$ as its argument changes; they require no exchange of the L and μ limits.

5. Numerical diagnostics

The table below compares direct finite-lattice Fourier integration of $\mathrm{Tr}(\Delta_L^2)$ with independent quadrature of the Bessel representation (2). Numerical values are rounded and are not interval enclosures.

μ	$H(\mu)$	$\|\Delta_{16}\|_2^2$	$\|\Delta_{64}\|_2^2$	$\|\Delta_{128}\|_2^2$
0.00625	0.00009107247233	0.00009059369729	0.00009097366525	0.00009102547241
0.05	0.002852564396	0.002817285774	0.002844634318	0.002848706782
0.5	0.084815500230	0.080385074026	0.083727553206	0.084275474084
2	0.471615679441	0.406078605117	0.454191601689	0.462834231123

As a separate check, the finite Frobenius coefficient sum was compared against explicit construction of the complex block Toeplitz matrix, and the matrix-entropy inequality was tested on noncommuting positive contractions. All derivations above are analytical; the numerical diagnostics help expose normalization and Fourier-sign mistakes.

References

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